

Name: Student ID:

ECS401: Unit 4 Lab Progress Test

This sheet is to help you and us judge your progress. You must complete it at the start of your lab session before any other activity (including getting work marked). Compare notes with others nearby and make sure everyone understands the right answers (available on the QM+ site). Talk to a demonstrator about anything that is at all unclear. Do NOT spend more than 15 minutes on answering the questions: if you don't know, leave it and get someone to explain.

1. **Complete the dry run table** and then **give** the final integer value of `z` printed to the screen.

```
int z = 0;                // L1
for(int j = 1; j <= 2; j++) // L2
{
    z = z + 1;            // L3
}
System.out.println(z);    // L4
```

Line	z	j	j <= 2 ?	printed to screen

Answer:

2. **Complete the dry run table** and then **give** the final integer value of `z` printed to the screen.

```
int z = 5;                // L1
for(int j = 1; j <= 4; j++) // L2
{
    z = z + 1;            // L3
}
System.out.println(z);    // L4
```

Line	z	j	j <= 4 ?	printed to screen

Answer:

3. **Complete the dry run table** and then **give** the final integer value of z printed to the screen.

```
int z = 1;                // L1
for(int k = 1; k < 1; k++) // L2
{
    z = z + 1;            // L3
}
System.out.println(z);    // L4
```

Line	z	k	k < 1 ?	printed to screen

Answer:

4. **Complete the dry run table** and then **give** the final integer value of z printed to the screen.

```
int z = 0;                // L1
for(int k = 1; k < 4; k++) // L2
{
    z = z + 2;            // L3
}
System.out.println(z);    // L4
```

Line	z	k	k < 4 ?	printed to screen

Answer:

5. **Complete the dry run table** and then **give** the final integer value of z printed to the screen.

```
int z = 0;                // L1
for(int j = 1; j <= 2; j++) // L2
{
    z = z + 1;            // L3
}
for(int k = 1; k <= 2; k++) // L4
{
    z = z + 1;            // L5
}
System.out.println(z);    // L6
```

Line	z	j	j <= 2 ?	k	k <= 2 ?	printed

Answer:

Terminology

6. A **for loop** is a counter-controlled loop: a programming construct that uses a counter to repeat code a fixed number of times. Highlight/underline each of the for loops in the following fragment of code.

```
int z = 0;
for(int j = 1; j <= 4; j++)
{
    z = z + 1;
}
for(int k = 1; k <= 4; k++)
{
    z = z + 1;
}
System.out.println(z);
```

7. The **loop body** is a block that is repeatedly executed under the control of the loop. Highlight/underline the loop body in the following fragment of code.

```
int z = 0;
for(int j = 1; j <= 4; j++)
{
    z = z + 1;
}
```

8. A **loop counter** is a variable that keeps track of how many times a loop body has executed. Highlight/underline the loop counter in the following fragment of code.

```
int z = 0;
for(int j = 1; j <= 4; j++)
{
    z = z + 1;
}
```

9. A **loop counter** must be **initialised**. Highlight/underline the statement that initialises the loop counter.

```
int z = 0;
for(int j = 1; j <= 4; j++)
{
    z = z + 1;
}
```

10. The termination of a **for loop** is controlled by a boolean expression. Highlight/underline the boolean expression in the following fragment of code.

```
int z = 0;
for(int j = 1; j <= 4; j++)
{
    z = z + 1;
}
```